Mohammad Al Olaimat

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Education

- **Ph.D. in Computer Science**
  University of North Texas (UNT), Denton, Texas, USA August 2020 - May 2025
  - Dissertation Title: " Modeling Electronic Health Records: Interpretable Sequential Approaches for Enhanced Multimodal Embeddings and Future Clinical Outcome Prediction with Time-Aware Irregular Intervals Handling"
  - Advisor: Dr. Serdar Bozdag
- **Master of Science in Computer Science**
  Texas A&M University-Commerce (TAMUC), Commerce, Texas, USA August 2017 - August 2019
  - Thesis Title: " Learning-based Data Augmentation for Multiclass Data"
  - Advisor: Dr. Jinoh Kim
- **Master of Science in Computer Science**
  Jordan University of Science and Technology (JUST), Irbid, Jordan February 2008 - May 2012
  - Thesis Title: " Job Migration in Contiguous Processor Allocation for Two-Dimensional Multicomputer"
  - Advisor: Dr. Wail Mardini
- **Bachelor of Science in Computer Science**
  AL al-Bayt University, Mafraq, Jordan August 2002 - June 2006

Research Interest

- Artificial Intelligence & Biomedical Informatics: Developing AI-driven computational methods for analyzing complex biomedical and clinical data, with applications in aging, Alzheimer's disease and related dementias (AD/ADRD), and precision medicine.
- Multimodal AI & Data Integration: Integrating electronic health records (EHRs), clinical notes, neuroimaging, molecular biomarkers, genetics, and omics data to investigate disease mechanisms, progression, and clinical outcomes.
- Computational Phenotyping & Real-World Health Data: Developing approaches for clinical data harmonization, intrinsic capacity phenotyping, information extraction, and longitudinal analysis of real-world healthcare data, particularly in aging and long-term care populations.
- Machine Learning & Deep Learning: Designing and applying advanced machine learning architectures, including Transformers, graph neural networks (GNNs), and recurrent neural networks (RNNs), for representation learning, predictive modeling, and high-dimensional biomedical data analysis.
- Computational Biology & Bioinformatics: Applying computational and AI-based approaches to genomic, transcriptomic, proteomic, and epigenomic data to investigate biological mechanisms and identify disease-associated biomarkers.

Publications

- Bruce J. Moeller, Michael Lozano Jr., Lindsay J. Peterson, **Mohammad Al Olaimat**, Mingyang Li, Alina Hagen, Hongdao Meng. "Projecting EMS Workforce Demand in an Aging State: A Florida Forecast Through 2035" medRxiv (2026): 2026-07. doi: https://doi.org/10.64898/2026.07.07.26357403
- Rahman, Most Tahmina, **Al Olaimat, Mohammad**, Bozdag, Serdar, and for Alzheimer's Disease Neuroimaging Initiative. "DyGraphTrans: A temporal graph representation learning framework for modeling disese progression from Electronic Health Records. " **bioRxiv** preprint. (2026) doi:10.64898/2026.01.28.702347. PMID: 41684944; PMCID: PMC12893049.
- **Al Olaimat, Mohammad,** Serdar Bozdag, and for Alzheimer's Disease Neuroimaging Initiative. "CAAT-EHR: Cross-Attentional Autoregressive Transformer for Multimodal Electronic Health Record Embeddings." **arXiv** preprint arXiv:2501.18891 (2025)
- Ozdemir, Cagri, **Mohammad Al Olaimat**, Serdar Bozdag, and Alzheimer's Disease Neuroimaging Initiative. "A Dynamic Model for Early Prediction of Alzheimer's Disease by Leveraging Graph Convolutional Networks and Tensor Algebra." In **Biocomputing** 2025: **Proceedings of the Pacific Symposium**, pp. 675-

689. 2025

- Susman, Aviad, Rupak Krishnamurthy, **Mohammad Al Olaimat,** Serdar Bozdag, Bino Varghese, Nasim Sheikh-Bahaei, and Gaurav Pandey. "Longitudinal ensemble integration for sequential classification with multimodal data." In 2025 **IEEE International Conference on Digital Health (ICDH),** pp. 31-40. **IEEE,** 2025.
- **Al Olaimat, Mohammad,** Serdar Bozdag, and Alzheimer's Disease Neuroimaging Initiative. "TA-RNN: an attention-based time-aware recurrent neural network architecture for electronic health records." **Bioinformatics** 40, no. Supplement_1 (2024): i169-i179. Proceedings of the 32nd Conference on Intelligent Systems for Molecular Biology (**ISMB 2024**). Acceptance rate: 18.2%
- Ozdemir, Cagri, **Al Olaimat, Mohammad**, Yashu Vashishath, Serdar Bozdag, and Alzheimer's Disease Neuroimaging Initiative. "IGCN: Integrative Graph Convolutional Networks for Multi-modal Data." **arXiv** preprint arXiv:2401.17612 (2024)
- **Al Olaimat, Mohammad**, Jared Martinez, Fahad Saeed, Serdar Bozdag, and Alzheimer's Disease Neuroimaging Initiative. "PPAD: a deep learning architecture to predict progression of Alzheimer's disease." **Bioinformatics** 39, no. Supplement_1 (2023): i149-i157. Proceedings of the 31st Annual Intelligent Systems for Molecular Biology and the 22nd Annual European Conference on Computational Biology (**ISMB/ECCB 2023**). Acceptance rate: 17.9%
- **Al Olaimat, Mohammad**, Dongeun Lee, Youngsoo Kim, Jonghyun Kim, and Jinoh Kim. "A learning-based data augmentation for network anomaly detection." In 2020 29th International Conference on Computer Communications and Networks (**ICCCN 2020**), pp. 1-10. **IEEE**, 2020. Acceptance rate: 28.9%

Professional Experience

- **Assistant Research Professor**
  University of South Florida, Tampa, Florida — June 2025 – Present
  Bellini College of Artificial Intelligence, Cybersecurity, and Computing | Center for Innovation, Technology, and Aging (CITA)
  - Conduct interdisciplinary research at the intersection of artificial intelligence, biomedical informatics, aging, and Alzheimer's disease and related dementias (AD/ADRD), developing computational methods to investigate health and disease trajectories.
  - Develop machine learning and multimodal AI approaches for integrating and analyzing clinical, imaging, biomarker, genetic, and omics data from large-scale biomedical research cohorts, including ADNI and HABS-HD.
  - Develop computational phenotyping and clinical data harmonization frameworks to derive multidomain intrinsic capacity measures from real-world electronic health records and longitudinal aging datasets.
  - Apply biomedical informatics, natural language processing, and AI methods to real-world long-term care data for clinical information extraction, data standardization, phenotyping, and health outcomes research.
  - Lead and contribute to interdisciplinary grant proposals focused on AI-enabled biomedical research, AD/ADRD, aging, and healthcare technology development.
  - Mentor graduate and undergraduate students in machine learning, deep learning, biomedical data analytics, and computational research methods.
- **Graduate Research Assistant**
  University of North Texas, Denton, Texas — August 2020 – May 2025 (including Teaching Fellow in Fall 2024)
  Conducted NIH-funded research in artificial intelligence, biomedical informatics, and Alzheimer's disease, with responsibilities including:
  - Developed deep learning models for Alzheimer's disease progression prediction using multimodal electronic health records (EHRs), including laboratory results, clinical notes, imaging, and genetic data.
  - Designed time-aware deep learning architectures to model irregular temporal intervals and longitudinal patient trajectories in EHRs.
  - Developed interpretable deep learning and representation learning approaches to extract meaningful patterns from complex clinical data and improve model transparency.
  - Collaborated on interdisciplinary research integrating clinical, genomic, and other omics data for disease prediction and biomedical discovery.
  - Undergraduate Research Mentorship (2021–2022): Mentored two undergraduate students on Alzheimer's disease prediction using EHRs, providing guidance on experimental design, data analysis, and manuscript preparation.

- NIH Grant Proposal Development (2022): Contributed to drafting and preparing an NIH grant proposal focused on integrating multimodal EHR and omics data for Alzheimer's disease prediction.

- **Graduate Research Assistant**
  Texas A&M University-Commerce, Commerce, TX May 2018 – May 2019
  - Collaborated with researchers at **Lawrence Berkeley National Laboratory** on high-performance computing (HPC) and network data analysis, utilizing deep learning techniques to enhance the detection and analysis of network anomalies
  - Conducted research funded by the Electronics and Telecommunications Research Institute (**ETRI**), applying deep learning Generative Adversarial Networks (GANs) for data augmentation to improve the detection and analysis of network anomalies

Teaching Experience

- **Teaching Fellow, Bioinformatics Algorithms**
  University of North Texas, Denton, Texas August 2024 – January 2025
  Taught courses on bioinformatics algorithms, emphasizing algorithmic problem-solving and their applications to biological data, including sequence alignment, motif searching, and gene expression analysis. The courses were cross listed across multiple departments, such as computer science, mathematics, and biology, and served both undergraduate and graduate students
  - Delivered comprehensive lectures and course materials
  - Graded exams and evaluated student projects, providing detailed feedback
  - Provided academic support and mentorship during office hours
- **Graduate Teaching Assistant**
  Texas A&M University-Commerce, Commerce, TX January 2018 – May 2018
  - Assisted Dr. Jinoh Kim in teaching two undergraduate and graduate courses: Computer Networking and Information Security
  - Trained students on the use of the machine learning software WEKA in the Information Security course
  - Led the network programming section in the Networking course, instructing students on Java for socket programming
  - Evaluated student performance through grading homework, quizzes, and exams, and provided constructive feedback to support their learning
- **Computer Science Instructor**
  Jordanian Ministry of Education, Mafraq, Jordan August 2006 – May 2015
  - Taught introductory programming, computer networks, databases, and image processing to high school students through lectures and interactive lessons
  - Provided hands-on training and practical exercises to reinforce learning and develop practical skills
  - Designed assessments and offered mentorship to support student progress and address challenges

Poster and Oral Presentations

- **Al Olaimat, M**., Bozdag, S., UNT Research Day, TA-RNN: an Attention-based Time-aware Recurrent Neural Network Architecture for Electronic Health Records, UNT, Denton, TX, United States of America. (2024)
- **Al Olaimat, M**., Bozdag, S., Saeed, F., AAIC Conference, TA-RNN: a Time-aware Attention-based Deep Learning Architecture to Predict Progression of Alzheimer’s Disease, Alzheimer's Association, Philadelphia, PA, United States of America. (2024)
- **Al Olaimat, M.**, Bozdag, S., UNT Research Showcase, TA-PPAD: a Time-aware Attention-based Deep Learning Architecture to Predict Progression of Alzheimer’s Disease, College of Engineering, UNT, Denton, TX, United States of America. (2024)
- **Al Olaimat, M.**, Bozdag, S., BioDiscovery Institute Seminar Series, Visit2Vec: Embedding method for electronic health record data, BioDiscovery Institute, Denton, TX, United States of America. (2023)
- **Al Olaimat, M.**, Martinez, J., Saeed, F., Bozdag, S., BioDiscovery Institute Advisory Board, PPAD: a deep learning architecture to predict progression of Alzheimer’s disease, BioDiscovery Institute, Denton, TX, United States of America. (2023)
- **Al Olaimat, M.**, Bozdag, S., BioDiscovery Institute Seminar Series, Prediction of Alzheimer's disease progression using LSTM, BioDiscovery Institute, Denton, TX, United States of America. (2021)
- **Al Olaimat, M.**, Kim, J., The A&M-Commerce Annual Research Symposium, A learning-based data

augmentation for network anomaly detection, TAMUC, Commerce, TX, United States of America. (2019)

Professional Service

- **Program Committee Member**, 7th International Workshop on Scalable Machine Learning for Health and Biomedical Data (**SMLHBD 2025**), held in conjunction with the IEEE International Conference on Bioinformatics and Biomedicine (**IEEE BIBM**) 2026
- **Program Committee Member**, 6th International Workshop on Scalable Machine Learning for Health and Biomedical Data (**SMLHBD 2025**), held in conjunction with the IEEE International Conference on Bioinformatics and Biomedicine (**IEEE BIBM**) 2025
- **Program Committee Member**, IEEE 5th Workshop on High Performance Computing, Big Data Analytics, and Integration for Multi-Omics Biomedical Data (**HPC-BOD 2024**), held in conjunction with IEEE International Conference on Big Data (**IEEE BigData**), 2024
- **Journal Reviewer**, Bioinformatics (Oxford University Press), 2026.
- **Reviewer**, 32nd and 31st Conference on Intelligent Systems for Molecular Biology (**ISMB**), 2025, 2024 and 2023
- **Reviewer**, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2023
- **Reviewer**, IEEE International Conference on Bioinformatics and Biomedicine (**BIBM**), 2023 and 2022
- **Reviewer**, Workshop on Data Mining in Bioinformatics (**BIOKDD**), ACM SIGKDD Conference, 2023 and 2022
- **Reviewer**, International Workshop on High Performance Computing, Big Data Analytics, and Integration for Multi-Omics Biomedical Data (**HPC-BOD**), 2022
- **Reviewer**, Great Lakes Bioinformatics Conference (**GLBio**), 2021 (Conference Abstract)

Awards and Honors

- **Conference Travel Grants,** Alzheimer's Association International Conference (AAIC), BioDiscovery Institute and Department of Computer Science and Engineering, University of North Texas, 2024.
- **Best Poster Presentation Award,** BioDiscovery Institute Expo, University of North Texas, Denton, TX, 2024.
- **Summer Research Assistantship Awards,** BioDiscovery Institute, University of North Texas, 2021 and 2023.